Cross-larva consensus maps of motor- and luminance-correlated neurons in the larval zebrafish brain

Abstract

We pooled motor-correlated, motor-anticorrelated, dark- and light-correlated neurons from 22 larvae in atlas coordinates and kept cells passing a cross-individual nearest-neighbour test ($P < 0.025$). Retained motor-positive cells occupied hindbrain and cerebellum, dark cells tectum, pretectum and forebrain, and light cells forebrain, habenula and cerebellum. Robustness controls were not run, so 35–60% retention reflects clustering relative to random cells, not calibrated error.

Whole-brain calcium imaging in larval zebrafish lets single neurons be classified by how their activity relates to behaviour or stimuli. Comparing such neurons across animals, however, requires a common atlas and a rule for discarding cells whose locations are not reproducible. Légaré et al. imaged 22 Tg(elavl3:H2B-GCaMP6s) larvae by two-photon microscopy, registered them to mapZebrain and defined four populations per fish at $P < 0.01$: motor-positive (M+), motor-negative (M−), dark-correlated (D) and light-correlated (L) [1]. They then removed cells "located at coordinates of the brain where few other cells from the same functional category were found nearby in other animals" with a spatial P-value test ($P < 0.025$) [1], adapted from the cross-fish conservation screens of Chen et al. [2]. Here we reproduced that consensus step on the released per-fish masks. The aim was to show where each category occurs reproducibly across individuals.

We used the methods actually executed. For each fish we read the boolean masks `correlated_motor.npy`, `anticorrelated_motor.npy`, `correlated_dark.npy` and `correlated_light.npy`, together with `centroids_atlas` and `region_labels` from `data.hdf5`. Fish5, fish17 and fish20 have no motor files, so M+ and M− were pooled from 19 larvae and D and L from 22. The paper reports n = 18 tail-recorded larvae [1]; we could not explain the one-fish difference. For each category cell i we computed a nearest-neighbour distance to the same category in each other fish j, and D_i was the mean of these distances. To build the null, we drew the same number of cells uniformly from fish j's full segmented cell pool and averaged the resulting nearest-neighbour distances. We repeated this 1000 times (seed 20241030). A cell was retained if $p_i = (1 + \#\{\text{null} \le D_i\})/1001 < 0.025$. Region labels came from the 70-region mapZebrain parcellation [3]. Reading the region masks with y and z flipped reproduced the stored labels exactly for the fish checked. Retained cells outside every region are kept in the per-cell tables and flagged, but we do not interpret them anatomically. A reviewer spot-checked the results: D_i was recomputed for sampled L cells, and p-values and retention were recomputed for all L cells from the saved null. The data x axis is mirrored: in every image, image-left is the animal's true right.

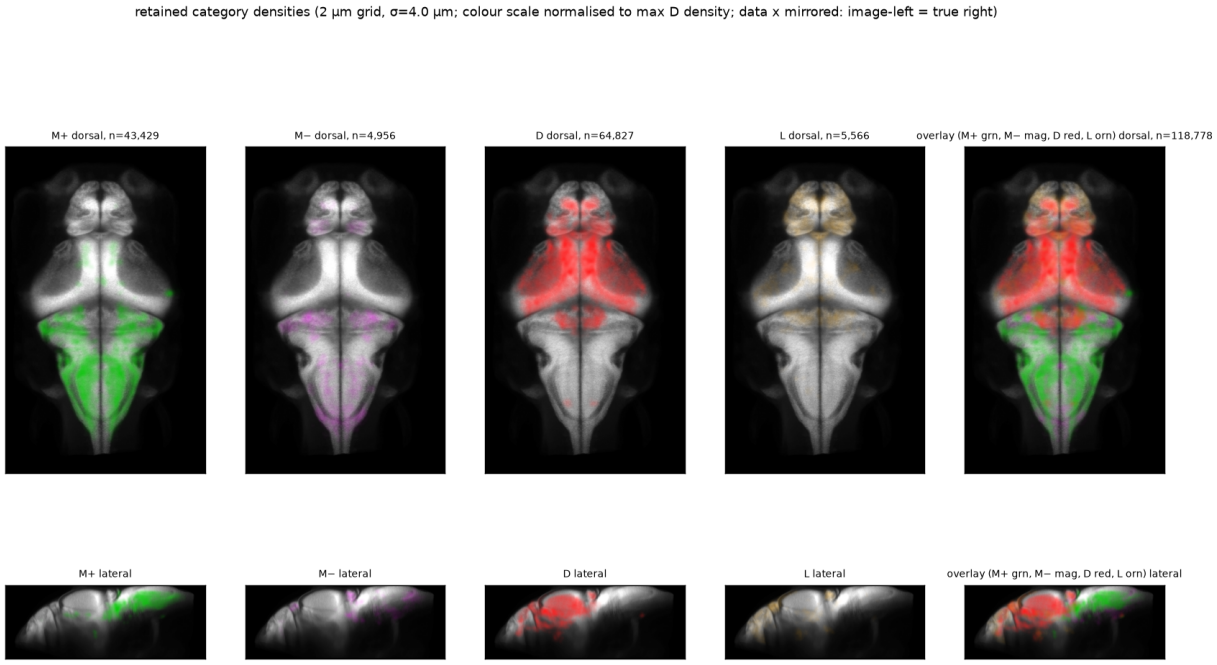


Figure 1. Kernel-density projections (2 µm grid, σ = 4 µm) of retained cells (spatial P < 0.025) for M+ (green, 19 fish), M− (magenta, 19 fish), D (red, 22 fish) and L (orange, 22 fish), on the mapZebrain H2B-GCaMP6s template; dorsal (top) and lateral (bottom) views, with a combined overlay at right. Colour is normalised to the maximum D density. Image-left = animal's true right.

The test retained 64,827 of 107,879 D cells (60.1%) and 43,429 of 79,415 M+ cells (54.7%). It retained 4,956 of 13,772 M− cells (36.0%) and 5,566 of 16,082 L cells (34.6%) (Table 1). Retention varied between fish, especially for the sparse categories: per-fish fractions ranged from 0.14 to 0.68 for M− and from 0.12 to 0.52 for L. The retained maps separate clearly (Figure 1). M+ consensus cells formed a broad, bilateral hindbrain field. The largest counts were in the inferior dorsal medulla stripes (InfDMOS2-3: 5,261; InfDMOS4: 4,721; InfDMOS5: 4,112), the medial octavolateralis nucleus (4,702), the vagus motor nucleus (3,635) and the cerebellum (2,575). In the facial motor nucleus and vagal sensory lobe, 86–99% of M+ cells were retained. No M+ cells were retained in the tectal periventricular layer (0 of 1,688) or the habenulae. M− consensus cells were sparser and more fragmented. They lay in the cerebellum (1,308), inferior medulla stripes and ventral medulla (InfDMOS2-3: 420; InfVMOR: 305), the nucleus isthmi (264) and the habenulae (DH 121, VH 204). All 48 M− cells in the locus coeruleus were retained, from 17 of 19 fish.

Table 1. Pooled counts before and after the spatial test, with selected regional retained counts (number of contributing fish in parentheses).

	M+ (19 fish)	M" (19 fish)	D (22 fish)	L (22 fish)
Cells before → retained	79,415 → 43,429	13,772 → 4,956	107,879 → 64,827	16,082 → 5,566
Pooled fraction retained	0.55	0.36	0.60	0.35
Retained, no atlas region (flagged)	3,787	583	1,712	305

	M+ (19 fish)	M" (19 fish)	D (22 fish)	L (22 fish)
Regions (of 70) with retained cells	33	31	41	37
Tectum PVL	0	4 (3)	26,220 (22)	316 (22)
Pretectum / dorsal thalamus (PTAP + DTP)	735	0	15,891 (22)	70
Dorsal + ventral habenula	0	325 (≤19)	2,194 (22)	1,037 (22)
Dorsal telencephalon (pallium)	8	10	2,245 (22)	921 (22)
Cerebellum	2,575 (19)	1,308 (19)	1,717 (22)	838 (22)
Locus coeruleus	112 (18)	48 (17)	0	0
Inf. dorsal medulla stripes 2–5	14,094 (19)	761 (≤19)	0	9
Vagus motor nucleus / vagal sensory lobe	4,915 (19)	367 (≤19)	0	0

Source: `tasks/core_consensus/counts_pooled.csv`, `regional_summary_long.csv`, `regional_consensus_fraction.csv`. Values marked "≤19" sum two regions whose contributing-fish counts differ.

The visual categories occupied mainly anterior territory. D consensus cells dominated the tectal periventricular layer (26,220; 85% of D cells there were retained) and the alar pretectum and dorsal thalamus (15,891). They also filled the superficial tectal strata (SFGS–SAC, 64–98% retained) and the pallium, subpallium and habenulae. No D cells were retained in the caudal medulla stripes, MON or vagal nuclei, even though roughly 120–1,800 D cells per region existed there before thresholding. L consensus cells were most numerous in the pallium (921), the cerebellum (838), the habenulae (1,037) and the subpallium (484). Smaller groups lay in the torus longitudinalis (418), olfactory bulb (385) and interpeduncular nucleus (103 of 121). Only 18% of L cells in the tectal periventricular layer survived. The categories overlapped in the cerebellum, which held retained cells of all four types.

Comparing unthresholded with retained densities shows what the test removed (Figure 2). Before thresholding, M+ cells were spread through the tectum and forebrain, D cells through the medulla, and L cells diffusely through the brain. These scattered cells were the ones removed. The anatomically compact cores remained, which makes the posterior-motor and anterior-visual segregation sharper.

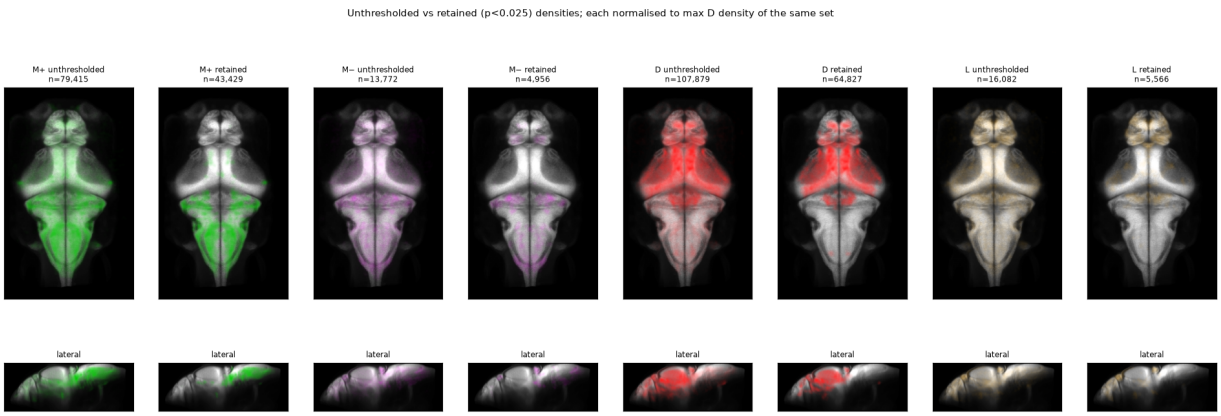


Figure 2. Unthresholded versus retained ($P < 0.025$) densities for each category, dorsal and lateral projections; each panel pair is normalised to the maximum D density of the same set. Image-left = animal's true right.

These maps reproduce the qualitative picture reported for this dataset. Motor-positive neurons lie broadly in the hindbrain. Motor-negative neurons form small segregated clusters in the hindbrain, cerebellum and habenula, and dark and light neurons concentrate in the tectum, cerebellum, habenula and forebrain [1]. The full M− retention in the locus coeruleus agrees with Légaré et al.'s overlay of M− clusters on noradrenergic markers [1]. That suggests, but does not show, that some quiescence-correlated cells are noradrenergic. The test establishes only spatial reproducibility of correlation with a regressor, not cell identity. A strong D consensus in the tectum and pretectum is consistent with the tectal OFF responses and whole-field dark/bright maps that Chen et al. reported [2]. Correlation alone cannot separate sensory, premotor or modulatory roles. Some M+ hindbrain cells may also be motor neurons or reticulospinal relays rather than pattern-generating circuitry.

Légaré et al. reported that light-responsive neurons concentrate in the left habenula [1]. In our retained L cells within the dorsal and ventral habenula, 575 lay at data $x > 290$ and 462 at $x < 290$. Because the x axis is mirrored, this is a modest excess on the animal's true left (55%). D habenular cells showed a similar excess (1,316 vs 878). These are pooled descriptive counts. The planned per-fish lateralisation test was not run, so we cannot claim a significant asymmetry. Hemisphere assignment also depends on the README's statement that x is mirrored.

The main limitation is statistical calibration. Between 35% and 60% of cells passed a nominal $P < 0.025$ threshold. This happened because the uniform whole-cell-pool null treats random neurons as the reference, and any spatially clustered category will beat it. "Retained" therefore means "closer to same-category cells in other larvae than random neurons are". It is not a calibrated false-positive rate, and it does not rank regions by reproducibility. The plan included several checks: leave-one-fish-out stability, random-mask null calibration, a null restricted to imaged tissue, threshold sensitivity and a habenula lateralisation test. None was executed, because the workflow skipped these dependent tasks, so the maps are unvalidated against those controls. Other limitations remain. The motor sample size differs from the published one (19 vs 18) for reasons we could not identify. Per-fish category sizes are uneven; for example, fish22 contributes 12,638 M+ cells. Registration error is unquantified beyond visual alignment with the template. Between 305 and 3,787 retained cells per category fall outside any atlas region. The reviewer fully recomputed only the L category. The ventral-most brain was not imaged [1].

In summary, this analysis produces atlas-registered consensus densities and per-cell tables (`retained_cells_3d.csv`, `consensus_cells_*.csv`) for all four categories. They separate a caudal motor territory from an anterior visual territory, with the cerebellum and habenula as mixed zones. The next step is a calibrated null that preserves each category's clustering and the per-fish robustness controls. Only then should these maps be read as reproducible functional nuclei.

References

1. Légaré A, Lemieux M, Boily V, Poulin S, Légaré A, Desrosiers P, De Koninck P. Structural and genetic determinants of zebrafish functional brain networks. Preprint (2024); code: github.com/pdesrosiers/Legare-2024-zebrafish-brain-networks.
2. Chen X, Mu Y, Hu Y, Kuan AT, Nikitchenko M, Randlett O, Sompolinsky H, Engert F, Ahrens MB. Brain-wide organization of neuronal activity and convergent sensorimotor transformations in larval zebrafish. *Neuron* (2018).
3. Shainer I, Kuehn E, Laurell E, Al Kassar M, Mokayes N, Sherman S, Larsch J, Kunst M, Baier H. A single-cell resolution gene expression atlas of the larval zebrafish brain. *Science Advances* (2023).